

# WEBSITE STATUS CHECKER

This project is developed using Python to check whether a website is online or offline by analyzing its HTTP status code.

## Objectives

- To learn Python networking basics.
- To understand HTTP status codes.
- To check website availability.
- To improve beginner Python programming skills.

## Software Requirements

| Software          | Purpose              |
|-------------------|----------------------|
| Python            | Programming Language |
| Requests Library  | Send HTTP Requests   |
| VS Code / PyCharm | Code Editor          |

## Python Program

```
import requests

url = input("Enter website URL: ")

try:
    response = requests.get(url)

    print("HTTP Status Code:", response.status_code)

    if response.status_code == 200:
        print("Website is Online")

    elif response.status_code == 404:
        print("Page Not Found")

    elif response.status_code == 500:
        print("Server Error")

    else:
        print("Other Status Detected")

except requests.exceptions.MissingSchema:
    print("Invalid URL")

except requests.exceptions.ConnectionError:
    print("Connection Error")

except Exception as e:
    print("Error:", e)
```

## Sample Output

```
Enter website URL: https://google.com
HTTP Status Code: 200
Website is Online
```

## Common HTTP Status Codes

| Status Code | Meaning               |
|-------------|-----------------------|
| 200         | Website Working       |
| 301         | Redirected            |
| 403         | Forbidden             |
| 404         | Page Not Found        |
| 500         | Internal Server Error |
| 503         | Service Unavailable   |

## Conclusion

The Website Status Checker project helps beginners understand how websites communicate using HTTP protocols. It also improves Python programming and networking knowledge.

## Viva Questions

1. What is HTTP?
2. What is status code 200?
3. Why is requests library used?
4. What is exception handling?
5. What is a GET request?